

Math Thermo
class #12
02/26/2026

The temperature relation can be inverted for U , i.e., there is a function $\tilde{U} = \tilde{U}(T, N)$, such that

$$\tilde{U}(T(U, N), N) = U, \quad \forall (U, N) \in \text{dom}(T)$$

Thus, there is a function

$$\tilde{S}_{\text{eq}} = \tilde{S}_{\text{eq}}(T, N)$$

defined by

$$(12.1) \quad \tilde{S}_{\text{eq}}(T, N) := S_{\text{eq}}(\tilde{U}(T, N), N).$$

Definition (12.1): The statistical Helmholtz free energy is defined as

$$(12.2) \quad F(T, N) := \tilde{U}(T, N) - T \tilde{S}_{\text{eq}}(T, N),$$

for all $(T, N) \in \text{dom}(\tilde{U})$.

Theorem (12.2): For the present case (N non-interacting particles with internal energy U), it follows that

$$(12.3) \quad F(T, N) = -N k_B T \log(Z),$$

where the partition function satisfies

$$(12.4) \quad Z = P_0 \int_Q \exp\left(-\frac{u(\vec{q})}{k_B T}\right) d\vec{q}.$$

Similar to before,

$$(12.5) \quad \tilde{U}(T, N) = N k_B T^2 \frac{\partial}{\partial T} [\log(Z)]$$

and

$$(12.6) \quad \tilde{S}_{eq}(T, N) = N k_B \log(Z) + N k_B T \frac{\partial}{\partial T} [\log(Z)],$$

Proof: The proofs are similar to those for the discrete energy case.

The details are left for an exercise. |||

Ideal Gas

For a monotonic ideal gas, we take the per-particle energy to be

$$u(\vec{q}) = \frac{1}{2} m (v_1^2 + v_2^2 + v_3^2)$$

for particles in the cube $\Omega = (0, L)^3$. The volume is of course

$$V = \int_{\Omega} d\vec{x} = L^3.$$

The partition function determines the thermodynamic

variables in the system.

$$\begin{aligned} Z &= P_0 \iint_{\mathbb{R}^3 \Omega} \exp\left(\frac{-m|\vec{v}|^2}{2k_B T}\right) d\vec{x} d\vec{v} \\ &= P_0 V \left[\int_{\mathbb{R}} \exp\left(\frac{-mv^2}{2k_B T}\right) dv \right]^3 \\ (12.7) \quad &= P_0 V \left(\frac{2\pi k_B T}{m} \right)^{3/2}. \end{aligned}$$

Now, we need to compute $\log(Z)$. Recall that $[Z] = 1$.

Let $T_0 > 0$ be a reference temperature. Then define

$$(12.8) \quad v_0 := \sqrt{\frac{2\pi k_B T_0}{m}},$$

which has units of velocity. Then

$$\begin{aligned} Z &= P_0 V \left(\frac{2\pi k_B T_0}{m} \right)^{3/2} \left(\frac{T}{T_0} \right)^{3/2} \\ (12.9) \quad &= P_0 V v_0^3 \left(\frac{T}{T_0} \right)^{3/2}. \end{aligned}$$

Note

$$[P_0 V v_0^3] = 1 = \left(\frac{T}{T_0} \right)^{3/2}$$

Thus,

$$(12.10) \quad \log(Z) = \log(P_0 V v_0^3) + \frac{3}{2} \log\left(\frac{I}{T_0}\right).$$

Here, we have been careful to make the arguments of the log dimensionless.

$$(12.11) \quad \frac{\partial}{\partial T} \log(Z) = \frac{3}{2} \frac{T_0}{T} \cdot \frac{1}{T_0} = \frac{3}{2T}.$$

It follows that

$$(12.12) \quad \begin{aligned} F &= -N k_B T \log(Z) \\ &= -N k_B T \log(P_0 V v_0^3) \\ &\quad - \frac{3 N k_B T}{2} \log\left(\frac{I}{T_0}\right), \end{aligned}$$

$$(12.13) \quad \begin{aligned} U &= N k_B T^2 \frac{\partial}{\partial T} [\log(Z)] \\ &= \frac{3 N k_B T}{2}, \end{aligned}$$

and

$$(12.14) \quad \begin{aligned} S &= N k_B \log(Z) + N k_B T \frac{\partial}{\partial T} [\log(Z)] \\ &= N k_B \left\{ \log(P_0 V v_0^3) + \frac{3}{2} \log\left(\frac{I}{T_0}\right) \right\} \\ &\quad + \frac{3 N k_B}{2}. \end{aligned}$$

For simplicity, we have dropped tildes, etcetera.

We can compute the pressure via

$$P = - \frac{\partial F}{\partial V}.$$

Recall that, in general, we have

$$dF = -SdT - PdV + \mu dN.$$

Thus,

$$P = \frac{Nk_B T}{V}.$$

It follows that

(12.15)

$$PV = Nk_B T.$$

This is the familiar equation of state for an ideal gas.

Let us examine the Maxwell-Boltzmann distribution for the ideal gas. Recall,

$$f(\vec{x}, \vec{v}) = f^{\text{eq}}(\vec{x}, \vec{v}) = P_0 \frac{N}{Z} \exp\left(\frac{-m|\vec{v}|^2}{2k_B T}\right),$$

where

$$Z = P_0 V \left(\frac{2\pi k_B T}{m}\right)^{3/2} = P_0 V v_0^3 \left(\frac{T}{T_0}\right)^{3/2}.$$

Thus,

$$f^{eq}(\vec{x}, \vec{v}) = M(\vec{v}) = \frac{N}{V \left(\frac{2\pi k_B T}{m} \right)^{3/2}} \exp\left(\frac{-m|\vec{v}|^2}{2k_B T} \right).$$

The function M is called the Maxwellian.

What is the average velocity of particles in our ideal gas?

What would you guess to be the answer?

$$\begin{aligned} \frac{1}{V_{ave}} &:= \iint_Q \frac{f^{eq}(\vec{q})}{N} \vec{v} d\vec{q} \\ &= \int_{\mathbb{R}^3} \frac{\exp\left(\frac{-m|\vec{v}|^2}{2k_B T} \right)}{\left(\frac{2\pi k_B T}{m} \right)^{3/2}} \vec{v} d\vec{v} \\ &= \vec{0}. \end{aligned}$$

This makes sense, right?

Next, let us show that, as expected, the temperature is positive.

$$\begin{aligned} \frac{3Nk_B T}{2} &= U \\ &= \iint_Q f^{eq}(\vec{q}) u(\vec{q}) d\vec{q} \end{aligned}$$

$$\begin{aligned}
 &= \iint_Q M(\vec{v}) \overbrace{\frac{1}{2} m |\vec{v}|^2}^{\geq 0} d\vec{x} d\vec{v} \\
 &\geq 0
 \end{aligned}$$

Thus,

$$T \geq 0.$$